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EXPERIMENT 7

Aim: Prepare RMMM Plan.

Risk Mitigation, Monitoring and Management (RMMM) Plan

Project Title: AgriSmart – AI-powered Agriculture Assistance Platform

Prepared by: Gargi

Date: 20 April 2025

1. Objective

To ensure that the AgriSmart platform — which provides disease detection using a CNN model, multilingual support, chatbot assistance, fertilizer calculation, weather integration via APIs, and community features — remains robust, accurate, accessible, and secure throughout its development and deployment.

2. Introduction to RMMM

The RMMM plan addresses the identification, analysis, monitoring, and response to risks during the development and operational lifecycle of AgriSmart. It focuses on issues such as model inaccuracies, data dependency, integration failures, security loopholes, and user experience challenges, especially considering multilingual and rural user bases.

3. What is RMMM?

- **Risk Mitigation:** Actions to reduce the probability or impact of potential risks.
 - **Risk Monitoring:** Tracking known risks and identifying new ones throughout development.
 - **Risk Management:** Procedures to respond effectively when a risk occurs.
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4. Why This RMMM Plan is Needed



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Since AgriSmart is AI-driven, multilingual, and dependent on APIs and third-party services, a failure in any component can directly affect usability, trust, and outcomes for farmers. A tailored RMMM plan ensures we:

- Maintain high model accuracy for disease prediction
 - Keep APIs (weather, etc.) functional and resilient
 - Protect user data (especially in community forums and chatbot)
 - Deliver a consistent experience in both English and Hindi
 - Ensure accessibility for a largely non-tech-savvy user base
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5. Use Case: AgriSmart Overview

AgriSmart is a software platform that empowers farmers with:

- **Plant disease detection** using a trained CNN model
 - **Multilingual chatbot** support for agriculture-related queries
 - **Fertilizer recommendation calculator**
 - **Weather insights** using external APIs
 - **Community forums** for peer-to-peer interaction and knowledge sharing
 - **Support for English and Hindi**
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6. Benefits of RMMM in AgriSmart

- Reduces chances of misinformation due to AI errors
 - Ensures accessibility across different languages
 - Guards against data misuse or cyber-attacks
 - Maintains trust and reliability through stable API integration
 - Supports scalability and long-term adoption
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7. Challenges in Implementing RMMM

- Model accuracy may degrade over time (concept drift)
 - API limitations (rate limits, outages, outdated data)
 - Handling offensive or incorrect content in community/chatbot
 - UI/UX inconsistencies across languages
 - Limited digital literacy among end users
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8. RMMM Risk Information Table



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Risk ID	Date	Probability	Impact	Description	Mitigation / Monitoring	Management / Contingency Plan	Assigned To
AGS-01	20/04/2025	Medium (60%)	High	CNN model misclassifies plant diseases or performs poorly on unseen images	Use diverse dataset, include feedback loop, retrain periodically	Flag images with low confidence, show "Not sure" disclaimer, allow user to report wrong output	ML Lead
AGS-02	20/04/2025	High (75%)	Medium	Weather API is down or returns inaccurate data	Use fallback API provider, cache last known data	Notify users of outdated data, suggest manual verification	Backend/API Dev
AGS-03	20/04/2025	Medium (50%)	High	Users post harmful, misleading, or spammy content in community forums	Add content moderation, keyword filters, report/block system	Temporarily ban spammers, notify moderators, auto-hide flagged posts	Community Manager
AGS-04	20/04/2025	Medium (40%)	Medium	Chatbot fails to understand input in Hindi or responds incorrectly	Improve NLU model for Hindi, expand training corpus	Provide "Was this helpful?" feedback, fallback to FAQs	NLP/Chatbot Dev
AGS-05	20/04/2025	Low (30%)	High	Data breach or leakage of personal information from forums/chatbot	Use HTTPS, authentication, encrypted storage, no sensitive info stored	Notify affected users, follow data breach protocols	Security Lead
AGS-06	20/04/2025	High (80%)	Medium	Fertilizer calculator gives poor recommendations due to input error or wrong formula	Add input validation, show examples, get formulas validated by agronomist	Log edge cases, allow feedback, add expert correction pipeline	Backend/Domain Expert
AGS-07	20/04/2025	Low (20%)	Medium	English-Hindi translations create	Work with native language	Allow user to switch language on every screen easily	UI/UX Lead



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				confusion or lose meaning	reviewers, use contextual NLP translation models		

9. Detailed Risk Breakdown Example

Risk ID: AGS-01

Description:

The CNN model may incorrectly classify plant diseases, especially if lighting, background, or crop variety differs from training data.

Mitigation / Monitoring:

- Train on diverse, high-resolution datasets (multiple seasons, varieties)
- Include a "Not sure?" option when model confidence < 50%
- Set up a retraining pipeline based on user feedback and reported errors
- Monitor model performance using F1 score and confusion matrix

Management / Contingency Plan:

- If consistent error patterns are found, hotfix the model or disable affected categories temporarily
- Notify users that detection results are indicative, not prescriptive
- Provide links to nearby agriculture support centers or contact lines

Assigned To: ML Developer (CNN Specialist)

Conclusion:

Thus we successfully prepared a RMMM plan for IPD project.